Chapter 10 of ISLRv2:

#1. (Courtesy of Russell and Norvig) Suppose that a training set contains only a single example, repeated 100 times. In 80 of the 100 cases, the single output value is 1; in the other 20, it is 0. What will a neural network predict for this example, assuming that it has been trained on all training examples and reaches a global optimum? (Hint: To find the global optimum, differentiate the error function and set the resulting expression to zero.)

#2. Explain exactly why networks of perceptrons with linear activation functions are uninteresting (that is, networks of perceptrons where, for each perceptron, the output is some constant times the weighted sum of the inputs). Use equations if necessary.

#3. You are given n numbers and have to determine their mean. Develop a gradient descent rule for this purpose. Denote our current estimate of their mean as $\hat{x}$ and use the error function: $Error(\hat{x}) = 1/2\left(\sum_i \left(\hat{x} - x_i\right)\right)^2$.

#4. You are given the function $f(x, y) = x^2 + xy + y^2$ and are trying to find a local minimum using gradient descent. You randomly start with x = 1.3 and y = 5.4. Perform the first step of gradient descent with learning rate α = 0.01. Show the resulting values for x and y as well as all of your calculations.

#5. Is overfitting more or less likely when the training set is small or large? Is overfitting more or less likely when the number of parameters to learn (such as the number of weights in a neural network) is small or large?

#6. We will now build some neural networks to represent basic boolean functions. For simplicity, we use the threshold function as our basic units instead of the sigmoid function, where threshold(t) = +1 if the input is greater than 0, and 0 otherwise. We have inputs x_i and weights w_i. Suppose we are given boolean input data x_i where +1 represents TRUE and 0 represents FALSE. Also don't forget the "intercept" weight w_0.

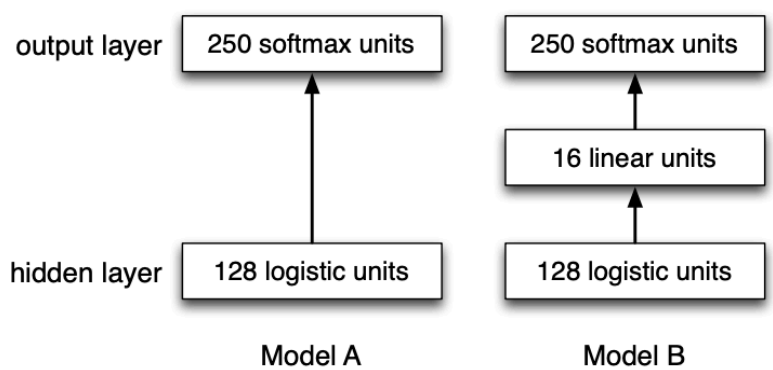
a) The boolean NOT function can be represented by a single-layer, single-unit neural net such that the output is +1 if and only if ¬x1 = TRUE. Choose the appropriate weights value for the neural net and draw a figure to represent the neural network.
b) Similarly like a), find a neural network with a single-layer, single-unit neural net to represent the OR function.
c) How about the AND function?

#7. Recall that the XOR function for two variables has the following truth table:

x_1	x_2	XOR
0	0	0
0	1	1
1	0	1
1	1	0

A neural net consisting of multiple perceptron units should be able to represent XOR function. Devise a multi-unit neural net (also multi-layer if needed) that computes the XOR of two inputs x1 and x2 using the same basic unit as the neural nets in #6. Draw the layout diagram and specify the weights of different edges clearly.
Hint: $XOR(x_1, x_2) = (Not(x_1)\ AND\ x_2)\ OR\ (Not(x_2)\ AND\ x_1)$

#8. Let's compare the following two models. Model A is the neural probabilistic language model. Model B is the same as Model A, with the following modification: in between the hidden layer (of size 128) and the output layer (of size 250), we insert a layer consisting of 16 linear units. The top layers of both models are shown in the figure. Describe one advantage of Model A and one advantage of Model B.



#9. Below is a diagram if a single artificial neuron (unit):

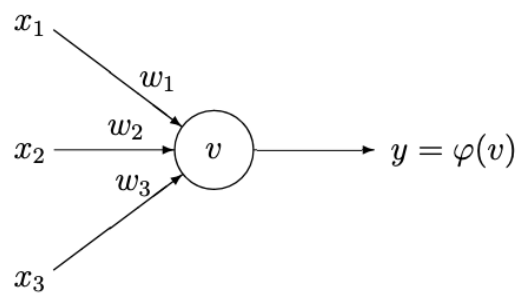


Figure 1: Single unit with three inputs.

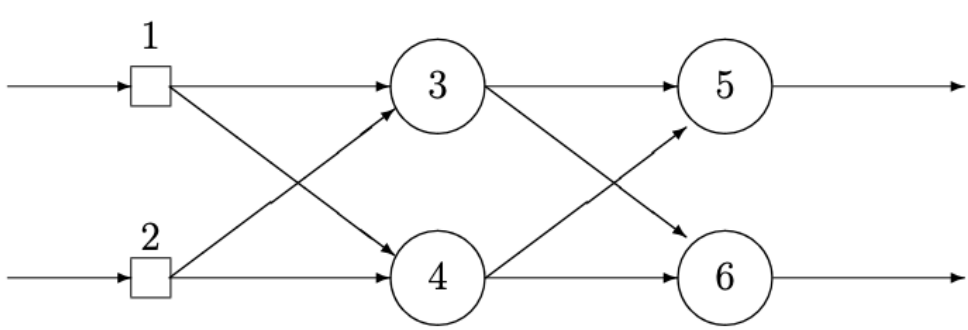
The node has three inputs $x = (x_1, x_2, x_3)$ that receive only binary signals (either 0 or 1). How many different input patterns this node can receive? What if the node had four inputs? Five? Can you give a formula that computes the number of binary input patterns for a given number of inputs?

#10. Consider the unit shown on Figure 1. Suppose that the weights corresponding to the three inputs have the following values: $w_1 = 2,\ w_2 =- 4,\ w_3 = 1$ and the activation of the unit is given by the step-function: $g(v) = 1,\ if\ v \geq 0;\ 0,\ otherwise$. Calculate what will be the output value y of the unit for each of the following input patterns:

Pattern	P_1	P_2	P_3	P_4
x_1	1	0	1	1
x_2	0	1	0	1
x_3	0	1	1	1

#11.

The following diagram represents a feed-forward neural network with one hidden layer:



A weight on connection between nodes i and j is denoted by w_{ij}, such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

$w_{13} = -2$	$w_{35} = 1$
$w_{23} = 3$	$w_{45} = -1$
$w_{14} = 4$	$w_{36} = -1$
$w_{24} = -1$	$w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

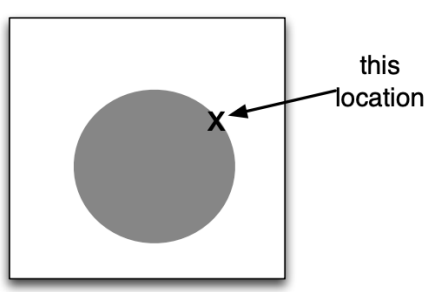
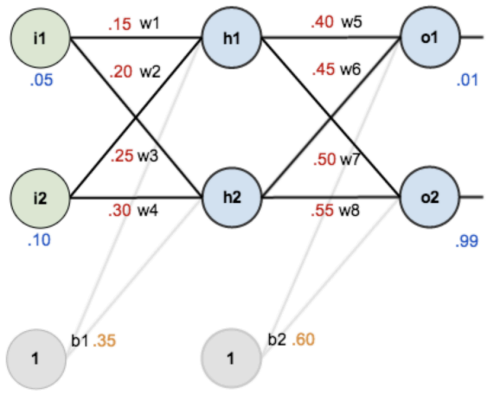
where v denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network (y_5 and y_6) for each of the input patterns:

Pattern:	P_1	P_2	P_3	P_4
Node 1:	0	1	0	1
Node 2:	0	0	1	1

#12. Suppose we compute the convolution I∗w, where I is a grayscale image (white pixels correspond to values of 1 and black pixels correspond to 0) and

$$w = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

is a convolution filter. For what parts of the image will the output be farthest from zero? In the image shown, will the output at the location marked by × take a positive or a negative value?



#13. A Neural Network applies sigmoid activation in the hidden layer as well as the output layer, and has initial weights, biases and training inputs/outputs as shown in the figure below. Use the back propagation algorithm with learning rate 0.5, perform a forward pass and a backward pass to update all the weights, then find out the updated prediction error.